# Singular Value Decomposition

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# Singular Value Decomposition (SVD) Method

- SVD is a factorization of a real or complex matrix.
- It generalizes the eigendecomposition of a square normal matrix with an orthonormal eigenbasis to any M*N matrix.

$$\overleftrightarrow{x} = \overleftrightarrow{U} \cdot \overleftrightarrow{\Sigma} \cdot \overleftrightarrow{V}^T$$

where $\overleftrightarrow{U}$ and $\overleftrightarrow{V}$ are unitary matrices—i.e., $\overleftrightarrow{U}^T = \overleftrightarrow{U}^{-1}$ and $\overleftrightarrow{V}^T = \overleftrightarrow{V}^{-1}$, so $\overleftrightarrow{V} \cdot \overleftrightarrow{V}^T = \overleftrightarrow{V} \cdot \overleftrightarrow{V}^{-1} = \overleftrightarrow{I}$ and the same for $\overleftrightarrow{U}$, hence their columns form orthonormal bases; and $\overleftrightarrow{\Sigma}$ is diagonal with the “singular values” of $\overleftrightarrow{X}$ in descending order.

# Singular Value Decomposition (SVD) Method

- The diagonal entries $\sigma_i = \Sigma_{ii}$ of $\Sigma$ are uniquely determined by M and are known as the singular values of M.
- The number of non-zero singular values is equal to the rank of x.

$$\overleftrightarrow{x} = \overleftrightarrow{U} \cdot \overleftrightarrow{\Sigma} \cdot \overleftrightarrow{V}^T$$

# SVD

Animated illustration of the SVD of a 2D, real shearing matrix $\mathbf{M}$. First, we see the unit disc in blue together with the two canonical unit vectors. We then see the actions of $\mathbf{M}$, which distorts the disk to an ellipse. The SVD decomposes $\mathbf{M}$ into three simple transformations: an initial rotation $\mathbf{V}^*$, a scaling $\mathbf{\Sigma}$ along the coordinate axes, and a final rotation $\mathbf{U}$. The lengths $\sigma_1$ and $\sigma_2$ of the semi-axes of the ellipse are the singular values of $\mathbf{M}$, namely $\mathbf{\Sigma}_{1,1}$ and $\mathbf{\Sigma}_{2,2}$.

https://en.wikipedia.org/wiki/Singular_value_decomposition

# SVD: EXAMPLE

Singular values can be considered as the importance values of different attributes in the data matrix.

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{U} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$\mathbf{\Sigma} = \begin{bmatrix} 3 & 0 & 0 & 0 & \mathit{0} \\ 0 & \sqrt{5} & 0 & 0 & \mathit{0} \\ 0 & 0 & 2 & 0 & \mathit{0} \\ 0 & 0 & 0 & \mathbf{0} & \mathit{0} \end{bmatrix}$$

$$\mathbf{V}^* = \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ -\sqrt{0.2} & 0 & 0 & 0 & -\sqrt{0.8} \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -\sqrt{0.8} & 0 & 0 & 0 & \sqrt{0.2} \end{bmatrix}$$

https://en.wikipedia.org/wiki/Singular_value_decomposition

# SVD: EXAMPLE

$$\mathbf{V}^* = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \sqrt{0.2} & 0 & 0 & 0 & \sqrt{0.8} \\ \sqrt{0.4} & 0 & 0 & \sqrt{0.5} & -\sqrt{0.1} \\ -\sqrt{0.4} & 0 & 0 & \sqrt{0.5} & \sqrt{0.1} \end{bmatrix}$$

# SVD

We know that

$$\overleftrightarrow{C} = \overleftrightarrow{V}\overleftrightarrow{\Lambda}\overleftrightarrow{V}^T$$

Then $\overleftrightarrow{V} = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_p \\ | & | & & | \end{bmatrix}$ and $\overleftrightarrow{\Lambda} = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_p \end{bmatrix}$,

with $|\lambda_1| > |\lambda_2| > \cdots > |\lambda_p|$.

**Relationship between the eigenvalues of covariance matrix and Singular Values in SVD**

- The columns of V are the principal directions
- Please note that covariance Matrix=XT.X/(n-1)

$$\overleftrightarrow{C} = \frac{1}{n-1}\overleftrightarrow{x}^T\overleftrightarrow{x}$$

$$= \frac{1}{n-1}\overleftrightarrow{V}\overleftrightarrow{\Sigma}\overleftrightarrow{U}^T\overleftrightarrow{U}\overleftrightarrow{\Sigma}\overleftrightarrow{V}^T$$

$$= \frac{1}{n-1}\overleftrightarrow{V}\overleftrightarrow{\Sigma}^2\overleftrightarrow{V}^T$$

$$= \overleftrightarrow{V}\left(\frac{\overleftrightarrow{\Sigma}^2}{n-1}\right)\overleftrightarrow{V}^T$$

$$\frac{\Sigma_{i,i}^2}{n-1} = \lambda_i$$

$U^TU=I$

Then $\overleftrightarrow{V} = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_p \\ | & | & & | \end{bmatrix}$ and $\overleftrightarrow{\Lambda} = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_p \end{bmatrix}$,

with $|\lambda_1| > |\lambda_2| > \cdots > |\lambda_p|$.

$$\overleftrightarrow{C} = \overleftrightarrow{V}\overleftrightarrow{\Lambda}\overleftrightarrow{V}^T$$

The columns of **V** are principal directions.

Relationship between the eigenvalues of covariance matrix and singular values

# Calculating SVD

- The non-zero singular values of M (the diagonal entries of Σ) are the square roots of the non-zero eigenvalues of both $X^T.X$ and $X.X^T$

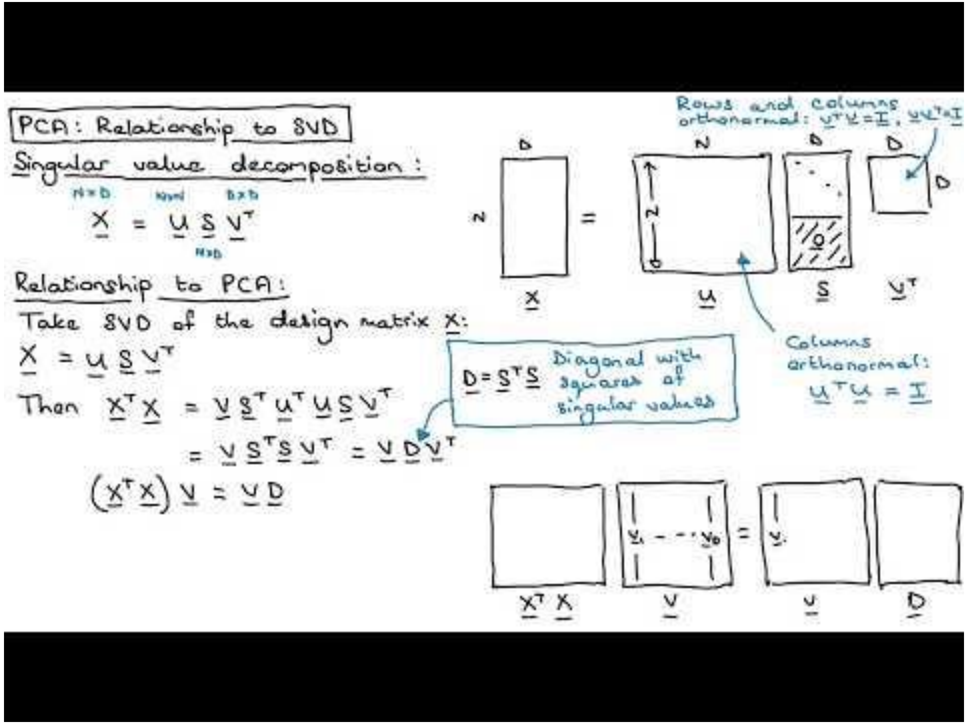
$$\frac{\Sigma_{i,i}^2}{n-1} = \lambda_i$$

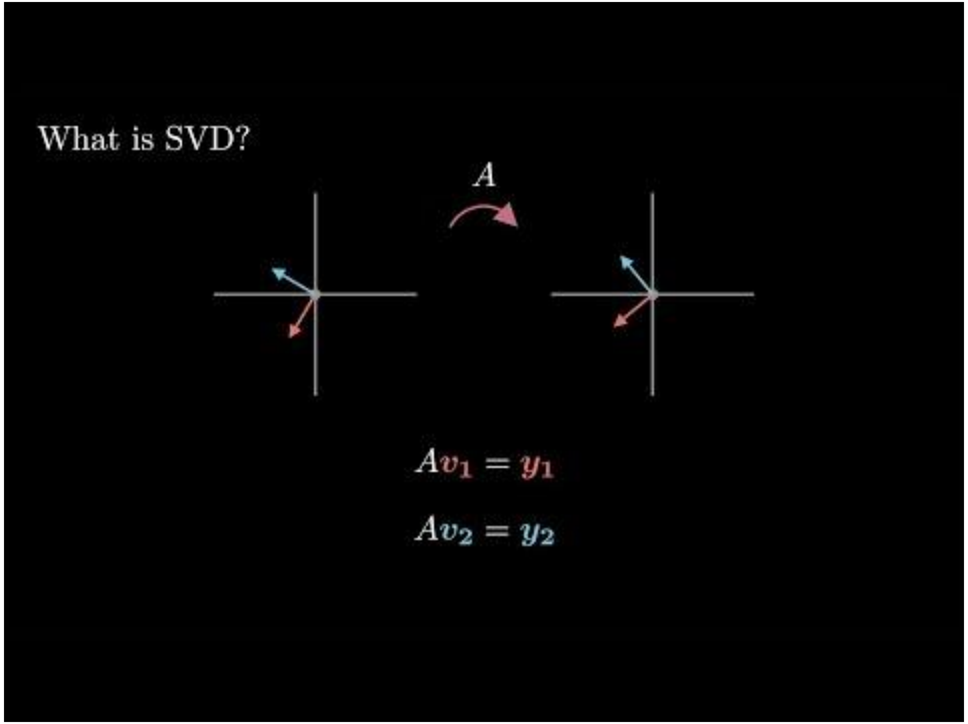
- The SVD of a matrix M can be computed by a two-step Numerical procedure

# Relationship between SVD and PCA

- Both dimensionality reduction
- Both PCA and SVD are methods of Matrix Decomposition
- You can calculate the PCA using SVD or using the eigendecomposition of the covariance matrix
- SVD also extracts data in the directions with the highest variances similar to PCA

# Relationship between SVD and PCA






https://www.youtube.com/watch?v=CpD9XlTu3ys

# SVD Example

https://www.d.umn.edu/~mhampton/m4326svd_example.pdf

# Movie Recommender System Using SVD

# User/Item Matrix


| | | | | |
|---|---|---|---|---|
| John  | 5 | 1 | 3 | 5 |
| Tom  | ? | ? | ? | 2 |
| Alice  | 4 | ? | 3 | ? |

https://www.researchgate.net/figure/Example-of-user-item-ratings-matrix-in-a-movie-recommendation-scenario_fig2_300646445

# Applying SVD on User/Item Matrix

M (5 x 5)

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 |
| 2 | 2 | 2 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 |
| 5 | 5 | 5 | 0 | 0 |
| 0 | 0 | 0 | 2 | 2 |

=

U (5 x 5)

| | | | | |
|---|---|---|---|---|
| 0.18 | 0 | 0.78 | 0.5 | 0.13 |
| 0.36 | 0 | 0 | 0.12 | 0.13 |
| 0.18 | 0 | 0 | 0 | 0 |
| 0.9 | 0 | 0 | 0 | 0 |
| 0 | 0.53 | 0.01 | 0 | 0 |

X

Σ (5 x 5)

| | | | | |
|---|---|---|---|---|
| 9.64 | 0 | 0 | 0 | 0 |
| 0 | 5.29 | 0 | 0 | 0 |
| 0 | 0 | 0.32 | 0 | 0 |
| 0 | 0 | 0 | 0.12 | 0 |
| 0 | 0. | 0 | 0 | 0.1 |

X

$V^T$ (5 x 5)

| | | | | |
|---|---|---|---|---|
| 0.58 | 0.58 | 0.58 | 0 | 0 |
| 0 | 0 | 0 | 0.71 | 0.71 |
| 0 | 0 | 0.8 | 0 | 0 |
| 0.6 | 0.8 | 0.1 | 0 | 0.8 |
| 0.54 | 0.54 | 0.03 | 0.2 | 0.8 |

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User Data

Product Data


U (5 x 2)

| | |
|---|---|
| 0.18 | 0 |
| 0.36 | 0 |
| 0.18 | 0 |
| 0.90 | 0 |
| 0 | 0.53 |

X

Σ (2 x 2)

| | |
|---|---|
| 9.64 | 0 |
| 0 | 5.29 |

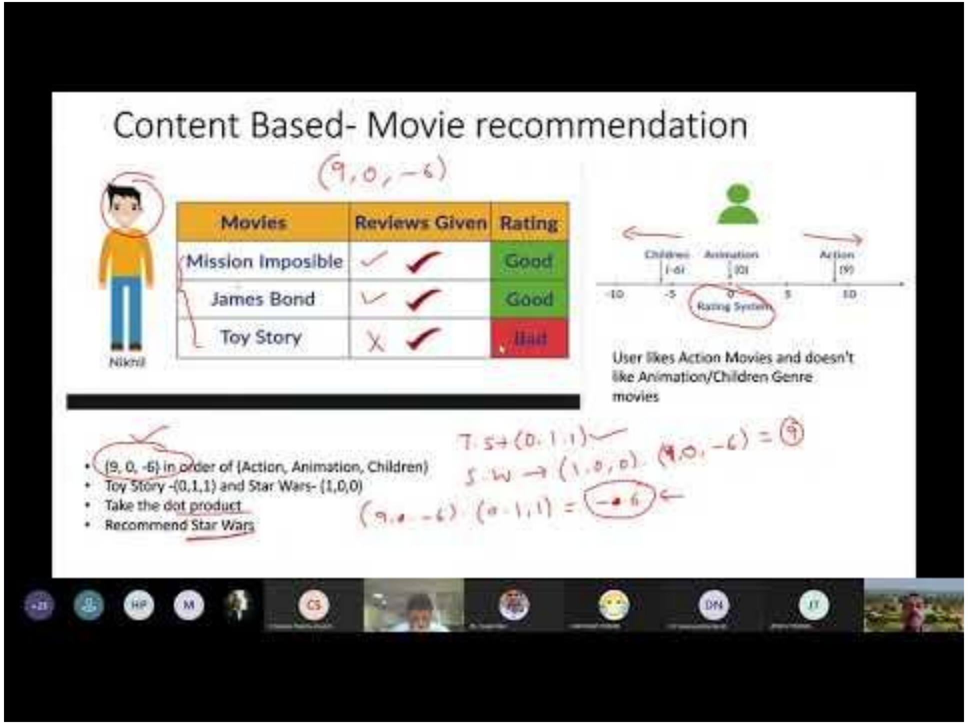
X

$V^T$ (2 x 5)

| | | | | |
|---|---|---|---|---|
| 0.58 | 0.58 | 0.58 | 0 | 0 |
| 0 | 0 | 0 | 0.71 | 0.71 |

https://www.mdpi.com/1996-1073/14/8/2284/htm

# Movie Recommendation Using SVD




https://www.youtube.com/watch?v=rFemvJgXY7E

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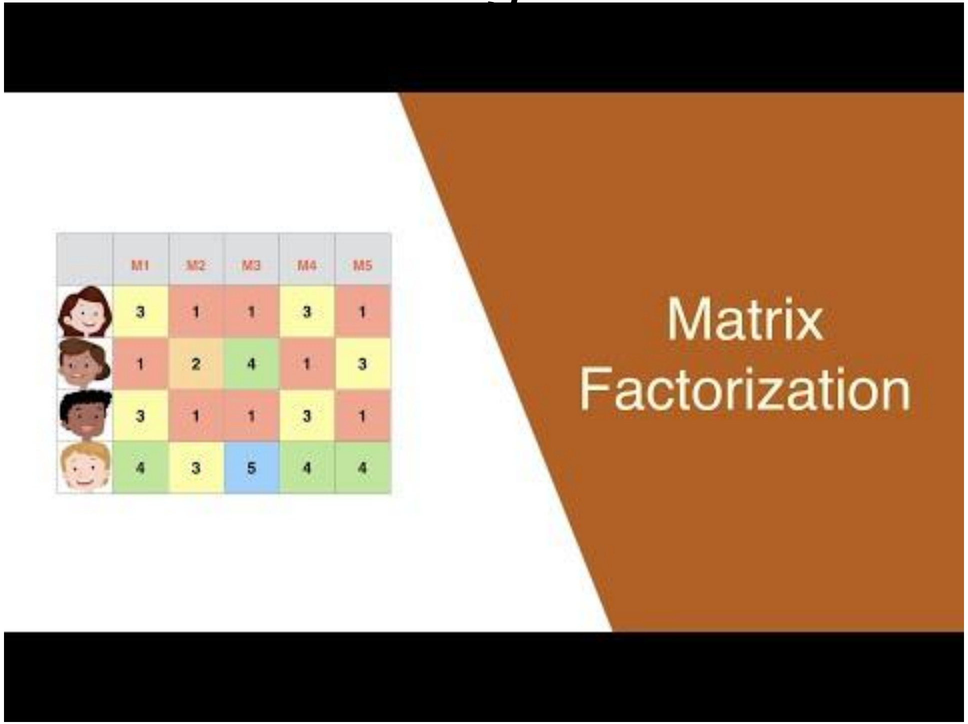
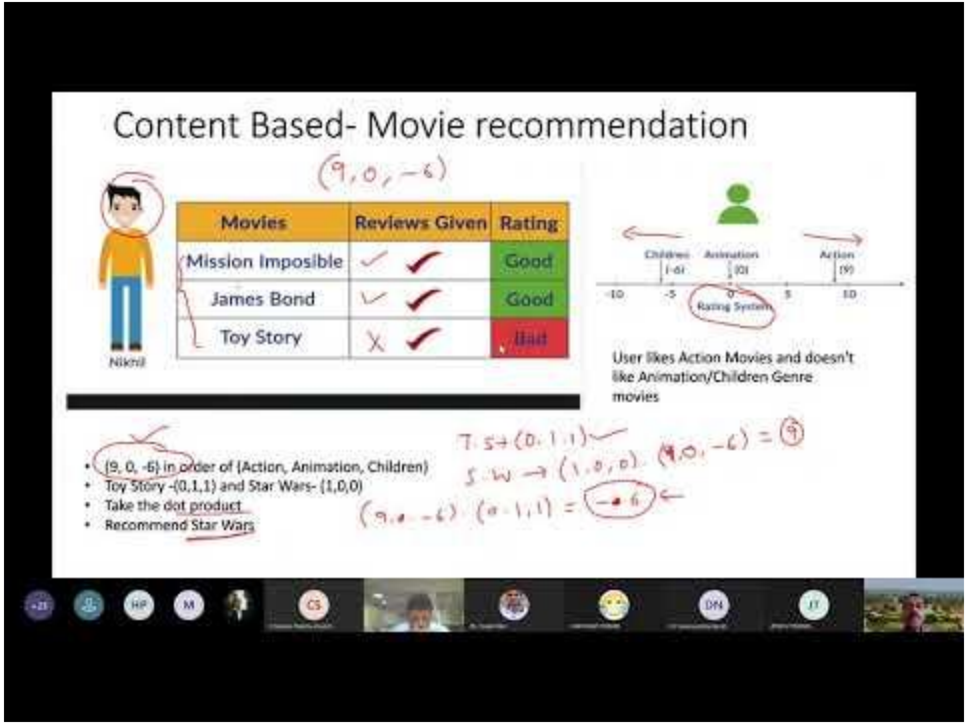



https://www.youtube.com/watch?v=ZspR5PZemcs




https://www.youtube.com/watch?v=rFemvJgXY7E

# Movie Recommendation Using SVD

https://towardsdatascience.com/beginners-guide-to-creating-an-svd-recommender-system-1fd7326d1f65